

Olist Brazilian Ecommerce

Sales Performance and Operational Analysis

A full analysis of order activity, regional performance, product categories, delivery outcomes, customer behaviour and payment patterns

Data period: September 2016 to August 2018 | Source: Public Olist dataset on Kaggle

1. Dataset Overview

This report is based on the Brazilian Ecommerce Public Dataset published by Olist. Olist operates as a large department store presence across multiple Brazilian marketplaces. The dataset contains real commercial transactions that have been anonymised. It covers approximately 100,000 orders placed between September 2016 and August 2018.

The data is organised across several related tables: orders, order items, customers, products, payments, reviews, sellers and a product category translation table. Together these tables allow analysis from multiple angles: commercial performance, geographic patterns, delivery reliability, product mix, payment behaviour and customer satisfaction.

Dataset link: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

2. Scope of Analysis

The analysis focused on delivered orders and examined the following areas:

- Overall commercial scale: total orders, revenue, average order value and unique customers
- Monthly sales trends across the observed period
- Revenue and order volume by customer state
- Top product categories by revenue and volume
- Delivery performance measured against estimated delivery dates
- Customer review score distribution
- Payment method mix and installment behaviour
- Share of one time buyers versus repeat customers
- Freight cost as a share of product price by region

All figures presented below come from SQL queries run on the full dataset after filtering to delivered orders where relevant. Results were consolidated in Excel for review and presentation.

3. Key Findings at a Glance

Total delivered orders	Product revenue (BRL)	Total GMV (BRL)	Avg order value (BRL)
96,478	13.59M	15.84M	164.20
Unique customers	Avg review score	One time buyers	Top state share
93,358	4.09 / 5	97.0%	SP ~42%

4. Detailed Findings

4.1 Regional concentration

Sao Paulo accounts for the largest share of both orders and revenue, contributing roughly 42 percent of total product revenue. Rio de Janeiro and Minas Gerais follow at a clear distance. This level of geographic concentration means overall results are sensitive to conditions that affect the Southeast, particularly Sao Paulo. Secondary states still contribute meaningful volume but remain far behind the top three.

4.2 Product categories

The strongest categories by revenue are bed bath table, health beauty, sports leisure, furniture decor and computers accessories. These categories consistently appear at the top of both revenue and item volume rankings. Watches and gifts stand out for higher average item prices, while telephony shows lower average prices but solid order counts. Protecting availability and seller quality in the leading categories is important for maintaining current revenue levels.

4.3 Delivery performance

In the South and Southeast, average delivery times often come in earlier than the estimated date, which is a positive signal for customer experience. In contrast, several states in the North and Northeast show higher rates of late deliveries and longer average times from purchase to receipt. Late delivery rates in some of these states exceed 20 percent. This gap is one of the clearer operational differences visible in the data and is likely to influence review scores and the willingness of customers in those regions to order again.

4.4 Customer behaviour and retention

Approximately 97 percent of customers placed only a single order during the observed period. Only a small fraction returned for a second purchase, and an even smaller group ordered three or more times. This pattern is common in marketplace data but represents a clear opportunity. Improving the rate of second purchases would raise customer lifetime value without requiring the same cost of acquiring entirely new buyers.

4.5 Payments and reviews

Credit card is the dominant payment method, and a large share of transactions use installments. This aligns with typical Brazilian ecommerce behaviour. Review scores are generally strong, with the majority of ratings at 4 or 5. The overall average sits near 4.1 out of 5. Where delivery performance is weaker, review scores tend to soften, reinforcing the link between logistics reliability and perceived quality.

4.6 Freight cost pressure

Freight as a percentage of product price is noticeably higher in more distant states. This affects both margin and the attractiveness of the offer to the customer. Uniform free shipping rules may not fit all regions equally well. Regional adjustments to shipping thresholds or partnerships could improve the balance between conversion and cost.

5. Implications and Recommended Focus Areas

The current profile supports a solid core business in the Southeast with good overall satisfaction scores. At the same time, three structural patterns stand out as priorities:

First, revenue concentration in Sao Paulo creates geographic risk. Expanding presence and demand in secondary states would diversify the revenue base.

Second, the very high share of one time buyers limits compounding growth. Even modest improvements in repeat purchase rates would have a material effect on lifetime value.

Third, delivery reliability outside the South and Southeast remains uneven. Closing that gap would support both satisfaction and the chance of repeat orders in currently weaker regions.

Addressing these three areas together would reduce concentration risk, improve unit economics through retention, and strengthen the experience in regions that today underperform on delivery.

6. Conclusion

The Olist data presents a clear picture of a marketplace with strong commercial activity, a concentrated geographic footprint, healthy review scores and a customer base that rarely returns for a second order. Delivery performance is solid in the core Southeast markets and weaker further away. Product revenue is led by a stable set of categories that reward consistent availability and quality.

The analysis highlights practical levers: regional diversification, retention of existing customers, and targeted improvement of logistics in states with elevated late delivery rates. Acting on these levers would support more balanced growth and a more resilient operating model over the next periods of activity.

Data source: Brazilian Ecommerce Public Dataset by Olist (Kaggle). Analysis performed with SQL and Excel on delivered order records.